12 Statistical Calculations

The 10bII+ allows you to enter data for one- and two-variable statistics easily. Once data is entered, you can use the statistical functions to calculate the following:

- Mean and standard deviation
- Regression statistics or a best fit
- Estimation and forecasting
- Weighted mean
- Summation statistics: n, Σx, Σx^2, Σy, Σy^2, and Σxy.

Table 12-1 Statistics keys

Keys	Description
↴ →M C STAT	Clear statistics memory.
x-value Σ+	Enter one-variable statistical data.
x-value ↴ Σ+ Σ-	Delete one-variable statistical data.
x-value INPUT *y*-value Σ+	Enter two-variable statistical data.
x-value INPUT *y*-value ↴ Σ+ Σ-	Delete two-variable statistical data.
RCL Σ+	Opens editor for reviewing or editing statistical data.
↴ 7 $\bar{x},\bar{y}$ ↴ K SWAP	Means of ***x*** and ***y***.
↴ 6 $\bar{x}_{w},b$ ↴ K SWAP	Mean of ***x*** weighted by ***y***. Also calculates ***b*** coefficient.
↴ 8 Sx,Sy ↴ K SWAP	Sample standard deviations of ***x*** and ***y***.
↴ 9 $\sigma x,\sigma y$ ↴ K SWAP	Population standard deviations of ***x*** and ***y***.
↴ 4 $\hat{x},r$ ↴ K SWAP	Estimation of ***x***. Also calculates ***r*** correlation coefficient.
↴ 5 $\hat{y},m$ ↴ K SWAP	Estimation of ***y***. Also calculates slope and ***m*** coefficient.
↵ REGR 1	Permits selection of six regression models or a best fit. Default is linear.

Clearing Statistical Data

Clear the statistical data before entering new data. If you don't clear the statistical data, new information stored will be added to the current calculations. To clear all statistical data, press [2nd] [C STAT]. The message **STAT CLR** flashes briefly and the display is cleared. The regression model is also reset to its default setting, **LINEAR**.

Entering Statistical Data

The 10bII+ uses a combination of list-based and register-based statistics when storing statistical data. List-based statistics store every value and permit you to review and edit entered data. Register-based statistics accumulate information, but you cannot easily edit or review this information.

On the 10bII+, there is always space reserved for up to 15 data points. In addition, up to 30 additional data points may be stored in memory shared with the cash flow memory. See Figure 1.

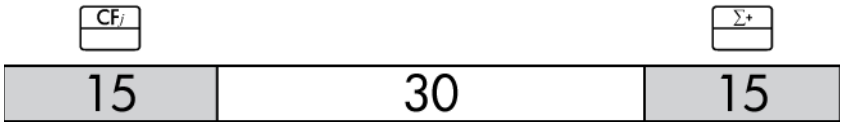


Figure 1

As illustrated in Figure 1, if no more than 15 cash flows are stored in the cash flow memory, you may store up to 45 data points for statistical usage.

If more than 15 cash flows are stored in the cash flow memory, the total memory available for storing statistical data is reduced. For example, in Figure 2, there are 25 cash flows stored, and the amount of available shared memory has therefore decreased by 10 slots.

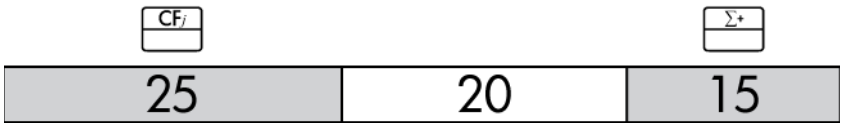


Figure 2

If data storage in the calculator memory resembles Figure 2, and you have a statistical calculation requiring more than 35 data points, clearing unneeded cash flow information will free up more space for information. If there are more data points than available memory, the 10bII+ automatically switches to register-based statistics to allow continued work. When available memory is reached, the **FULL** annunciator indicates there is not enough space to continue saving data. See Figure 3.

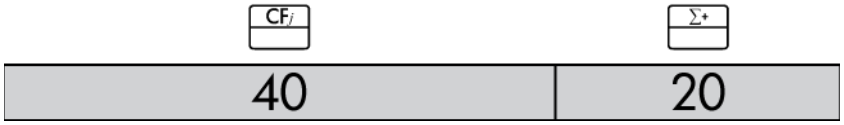


Figure 3

When the calculator switches to register-based mode, some key points to consider:

- You may enter an unlimited number of data points.
- The statistics editor, accessed with RCL Σ+, is not available.
- While use of ↘ Σ+/Σ- is allowed, viewing previously entered data is not possible.
- The only regression mode available is a linear regression.

One-Variable Statistics

To enter ***x***-data for one-variable statistics complete the following steps:

1. Clear the statistical registers by pressing ↘ →M/C STAT.
2. Enter the first value and press Σ+. The HP 10bII+ displays ***n***, the number of items accumulated.
3. Continue accumulating values by entering the numbers and pressing Σ+. The ***n***-value is increased with each entry.

Two-Variable Statistics and Weighted Mean

To enter ***x,y*** pairs of statistical data complete these steps:

1. Clear the statistical registers by pressing ↘ →M/C STAT.
2. Enter the first ***x***-value and press INPUT. The HP 10bII+ displays the ***x***-value.
3. Enter the corresponding *y*-value and press Σ+. The HP 10bII+ displays ***n***, the number of pairs of items accumulated.
4. Continue entering ***x,y*** pairs. The ***n***-value is increased with each entry.

To enter data for calculating the weighted mean, enter each data value as ***x***, and its corresponding weight as *y* in the statistics memory. Press ↘ 6/$\bar{x}_{w,b}$ to calculate the weighted mean.

Viewing and Editing Statistical Data

1. Press RCL Σ+ to open the editor. The number of items accumulated, n, is displayed, along with the current x-or y-value. The **STAT** annunciator appears, and the **X** or **Y** identifies the displayed value.
2. Press + to move up through the current statistical data. When you pass the maximum of the data, an empty statistical pair displays before wrapping back to x_1, provided there is enough memory for more data.
3. Press − to move down through the current statistical data. At x_0, the display wraps back to the maximum y-value.
4. At any time with the editor open, press Σ+ to return to x_1. To jump to a specific data pair, type the whole number which represents the pair's n-value and press Σ+. The editor will jump to that data pair, unless your entered number is higher than your maximum data pair, in which case it will jump to the highest x-value. If you type in an invalid number, such as a negative number, or a non-whole number, the editor remains in its current position.
5. To delete the currently displayed statistical data pair, press ÷. To add a new pair with the x- and y-values equal to zero, press ×.
6. To replace the currently displayed value, simply type in the new number and press INPUT.
7. To clear the currently displayed x-or y-value without removing the entire pair, press ← to set the value to 0.
8. Press C to exit the editor.

Example 1

A tropical beach resort has been having some very hot weather lately. A manager at the beach resort has noticed an increase in the number of cold drinks sold during hot days and wants to be able to predict how many employees are needed to sell drinks tomorrow. Each employee can sell 200 drinks a day at most.

Table 12-2 Data

Past 3 Days Temperature (Celsius)	Cold Drinks Sold
32	415
35	515
38	725

At what temperature would the manager predict to sell 800 drinks? How many employees will be needed for tomorrow's predicted temperature of 43°C?

Table 12-3 Example entering statistical data, opening the editor, and predicting

Keys	Display	Description
[↓] [→M / C STAT]	**STAT CLR** (message flashes briefly, then disappears)	Clears the statistics memory.
[3] [2] [INPUT] [4] [1] [5] [Σ+]	1.00	Enters first ordered pair.
[3] [5] [INPUT] [5] [1] [5] [Σ+]	2.00	Enters the second ordered pair.
[3] [8] [INPUT] [7] [2] [5] [Σ+]	3.00	Enters the third ordered pair.
[RCL] [Σ+]	1 32.00	Open the editor. Displays **X** annunciator.
[+] [+] [+] [+] [+]	3 725.00	Scroll and verify the data points, starting with the *x*-value of the first pair. The *y*-value of the third pair is displayed.
[C]		Exit the editor.
[↑] [REGR / 1] [4]	0.00	Set the regression model to power. **4–POWER** flashes briefly after [4] is pressed, then disappears.
[8] [0] [0] [↓] [4 / x̂,r]	39.49	Predict the temperature.
[↓] [K / SWAP] [↓] [= / DISP] [9]	.988080878	Display the correlation coefficient.
[4] [3] [↓] [5 / ŷ,m] [↓] [= / DISP] [2]	1,053.49	Predict the number of drinks sold tomorrow.
[÷] [2] [0] [0] [=]	5.27	Manager should have at least 6 employees at work tomorrow to cover the expected load.

Continuing with this example, modify this data by adding more points: two additional days of sales and their corresponding temperatures. The first day's temperature of 43°C resulted in the sale of 1,023 cold drinks. The next day's temperature at 37°C resulted in the sale of 685 drinks.

Table 12-4 Adding more data

Keys	Display	Description
4 3 INPUT 1 0 2 3 Σ+	4.00	Enters fourth ordered pair.
3 7 INPUT 6 8 5 Σ+	5.00	Enters fifth ordered pair.

After modifying the data, predict the next day's activity at a record 45°C.

Table 12-5 A new prediction

Keys	Display	Description
4 5 ↓ 5 ŷ,m	1,204.67	Predicts the drinks sold at 45°C. But is this the best fit?
↑ REGR 1 0	0.00	Sets regression mode to **0-BEST FIT**.
4 5 ↓ 5 ŷ,m	1,128.12	All regressions are calculated and **LINEAR** is selected as being a better fit than **POWER**. The result, 1128, is well within the limit of six employees.

Summary of Statistical Calculations

The **STAT** annunciator indicates that a statistical calculation was performed. Some functions return two values. In this instance, the **X** annunciator is displayed along with **STAT**. Press [SWAP] to see the second value. In this case, the **X** annunciator changes to a **Y**, indicating the second value is being displayed.

Table 12-6 Statistical calculations that return two values

Keys	Description	[SWAP] Displays:
[7 $\bar{x},\bar{y}$]	Arithmetic mean (average) of the x-values.	Mean (average) of the y-values if you entered y-data.
[8 Sx,Sy]	Sample standard deviation of the x-values. NOTE: The sample standard deviation assumes that the data is a sampling of a larger, complete set of data. The population standard deviation assumes that the data constitutes the entire population.	Sample standard deviation of the y-values if you entered y-data. NOTE: The sample standard deviation assumes that the data is a sampling of a larger, complete set of data. The population standard deviation assumes that the data constitutes the entire population.
[9 $\sigma x,\sigma y$]	Population standard deviation of the x-values. NOTE: The sample standard deviation assumes that the data is a sampling of a larger, complete set of data. The population standard deviation assumes that the data constitutes the entire population.	Population standard deviation of the y-values if you entered y-data. NOTE: The sample standard deviation assumes that the data is a sampling of a larger, complete set of data. The population standard deviation assumes that the data constitutes the entire population.
y-value [4 $\hat{x},r$]	Estimate of x for a given value of y.	Correlation coefficient. NOTE: The correlation coefficient is a number in the range -1 through +1 that measures how closely the data fits the calculated line. A value of +1 indicates a perfect positive correlation, and -1 indicates a perfect negative correlation. A value close to zero indicates the line is a poor fit.
x-value [5 $\hat{y},m$]	Estimate of y for a given value of x.	Coefficient m of the current regression.
[6 $\bar{x}_{w},b$]	Mean of the x-values weighted by the y-values.	Coefficient b of the current regression.

Mean, Standard Deviations, and Summation Statistics

You can calculate the mean ($\bar{x}$), sample standard deviation (S_x), and population standard deviation (σ_x), and summation statistics, n, Σx, and Σx^2 of x-data. For x,y data, you can also calculate the mean, sample standard deviation, and population standard deviation of the y-data and the summation statistics Σy, Σy^2, and Σxy.

Example 2

A yacht captain wants to determine how long it takes to change a sail. She randomly chooses six members of her crew, observes them as they carry out the sail change, and records the numbers of minutes required: 4.5, 4, 2, 3.25, 3.5, 3.75. Calculate the mean and sample standard deviation of the times. Also, calculate the root mean square, using the formula, $\sqrt{\sum x^2/n}$.

Table 12-7 Example calculating mean, standard deviation, and summation statistics

Keys	Display	Description
[↘] [→M / C STAT]	**STAT CLR** (message flashes briefly, then disappears)	Clears statistics memory.
[4] [•] [5] [Σ+]	1.00	Enters first time.
[4] [Σ+]	2.00	Enters second time.
[2] [Σ+]	3.00	Enters third time.
[3] [•] [2] [5] [Σ+]	4.00	Enters fourth time.
[3] [•] [5] [Σ+]	5.00	Enters fifth time.
[3] [•] [7] [5] [Σ+]	6.00	Enters sixth time
[↘] [7 / $\bar{x},\bar{y}$]	3.50	Calculates the mean.
[↘] [8 / Sx,Sy]	0.85	Calculates the sample standard deviation.
[↗] [Σx^2 / 7]	77.13	Displays Σx^2 .
[÷] [↗] [n / 4]	6.00	Displays n.
[=] [↘] [$\bar{x}$ / $\sqrt{x}$]	3.59	Calculates the root mean square.

The standard deviations calculated by [→] [8 Sx,Sy] and [→] [8 Sx,Sy] [→] [K SWAP] are the sample standard deviations. They assume that the data is a sampling of a larger, complete set of data. If the data constitutes the entire population, the true population standard deviations can be calculated by pressing [→] [9 σx,σy] and [→] [9 σx,σy] [→] [K SWAP].

Example 3

The coach has four new players on the team with heights of 193, 182, 177, and 185 centimeters and weights of 90, 81, 83, and 77 kilograms. Find the mean and population standard deviation of both their heights and weights, then sum the y-data.

Table 12-8 Example 3

Keys	Display	Description
[→] [→M C STAT]	**STAT CLR** (message flashes briefly, then disappears)	Clears statistics memory.
[1] [9] [3] [INPUT] [9] [0] [Σ+]	1.00	Enters height and weight of player 1.
[1] [8] [2] [INPUT] [8] [1] [Σ+]	2.00	Enters height and weight of player 2.
[1] [7] [7] [INPUT] [8] [3] [Σ+]	3.00	Enters height and weight of player 3.
[1] [8] [5] [INPUT] [7] [7] [Σ+]	4.00	Enters height and weight of player 4.
[→] [7 $\bar{x},\bar{y}$]	184.25	Calculates mean of heights (x).
[→] [K SWAP]	82.75	Displays mean of weights (y).
[→] [9 σx,σy]	5.80	Calculates population standard deviation for heights (x).
[→] [K SWAP]	4.71	Displays population standard deviation for weights (y).
[↑] [Σy 6]	331.00	Displays the total of the y-values.

Linear Regression, Estimation, and Regression Modes

Linear regression is a statistical method for estimation and forecasting. It is used to find a straight line that best fits a set of x,y data. There must be at least two different x,y pairs. The straight line provides a relationship between the x- and y-variables: $y = mx + b$, where m is the slope and b is the y-intercept.

Linear Regression. Calculate ***r*** (the correlation coefficient), ***m***, ***b***, and as follows:

1. Clear the statistical registers by pressing [↓] [→M / C STAT].
2. Enter the first ***x***-value and press [INPUT]. The ***x***-value is displayed.
3. Enter the corresponding ***y***-value and press [Σ+]. The HP 10bII+ displays ***n***, the number of pairs of items accumulated.
4. Continue entering ***x,y*** pairs. The ***n***-value is increased with each entry.
5. To display ***r***, the correlation coefficient, press [↓] [4 / $\hat{x}$,r] [↓] [K / SWAP].
6. To display ***m***, the slope, press [↓] [5 / $\hat{y}$,m] [↓] [K / SWAP].
7. To display ***b*** (the ***y***-intercept), press [↓] [6 / $\bar{x}_{w,b}$] [↓] [K / SWAP].

Linear Estimation. The straight line calculated by linear regression can be used to estimate a *y*-value for a given ***x***-value, or vice versa:

1. Enter the ***x,y*** data.
2. Enter the known ***x***-value or ***y***-value.
   - To estimate ***x*** for the given ***y***, enter the ***y***-value, then press [↓] [4 / $\hat{x}$,r].
   - To estimate ***y*** for the given ***x***, enter the ***x***-value, then press [↓] [5 / $\hat{y}$,m].

Example: 4

Ali's Azaleas advertises on a local radio station. For the past six weeks, the manager has kept records of the number of minutes of advertising that were purchased, and the sales for that week.

Table 12-9 Recording the number of minutes of the advertisements and sales

Week	Minutes of Advertising (*x*-values)	Sales (*y*-values)
1	2	1,400
2	1	920
3	3	1,100
4	5	2,265
5	5	2,890
6	4	2,200

What is the *y*-intercept, the slope, and the correlation coefficient?

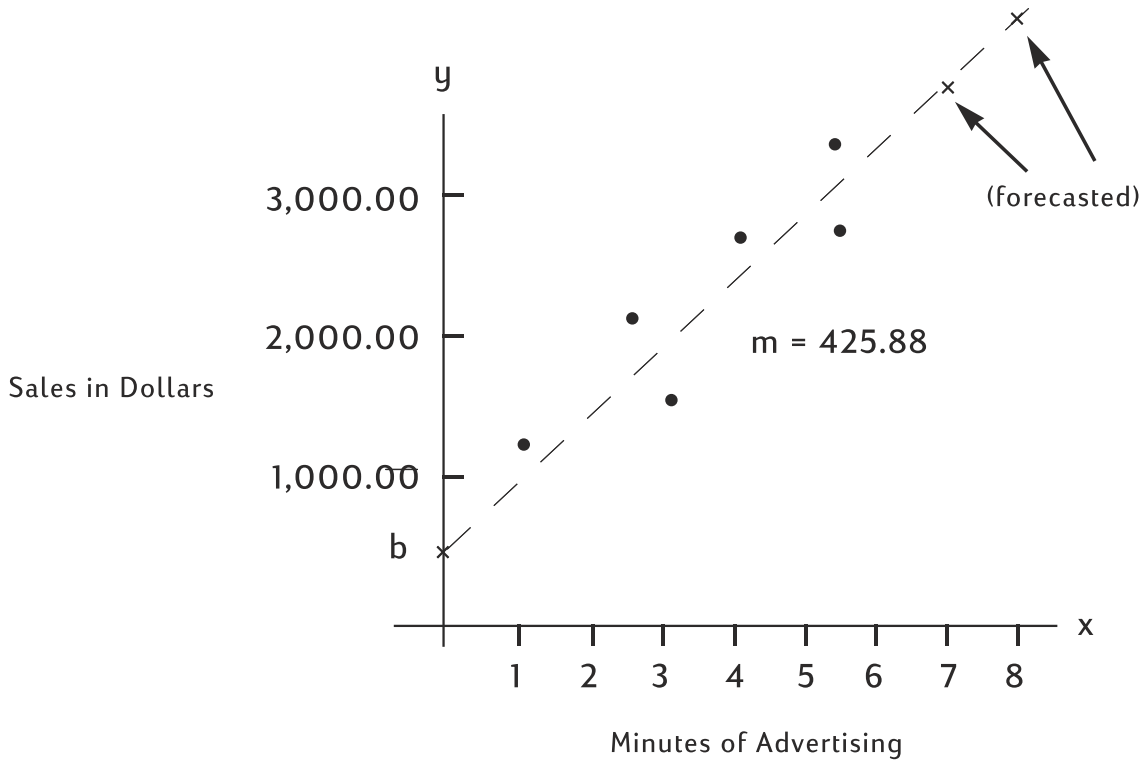


Figure 4 Diagram on forecasting sales and minutes for advertising

Table 12-10 Example for forecasting

Keys	Display	Description
↴ →M C STAT	**STAT CLR** (message flashes briefly, then disappears)	Clears statistics memory.
2 INPUT 1 4 0 0 Σ+	1.00	Enters minutes and sales for consecutive weeks.
1 INPUT 9 2 0 Σ+	2.00	
3 INPUT 1 1 0 0 Σ+	3.00	
5 INPUT 2 2 6 5 Σ+	4.00	
5 INPUT 2 8 9 0 Σ+	5.00	
4 INPUT 2 2 0 0 Σ+	6.00	
↴ 6 $\bar{x}_{w,b}$ ↴ K SWAP	376.25	Calculates ***y***-intercept.
↴ 5 $\hat{y},m$ ↴ K SWAP	425.88	Displays slope.

Table 12-10 Example for forecasting

Keys	Display	Description
[→] [4 $\hat{x}$,r] [→] [K SWAP]	0.90	Calculates correlation coefficient.

Estimate what the level of sales would be if the business purchased 7 or 8 minutes of advertising.

Table 12-11 Estimating the level of sales

Keys	Display	Description
7 [→] [5 $\hat{y}$,m]	3,357.38	Estimates sales if 7 minutes of advertising were purchased.
8 [→] [5 $\hat{y}$,m]	3,783.25	Estimates sales if 8 minutes were purchased.

How many minutes of advertising should Ali's buy to attain sales of 3,000?

Table 12-12 Estimating the minutes of advertising for 3,000 sales

Keys	Display	Description
3 0 0 0 [→] [4 $\hat{x}$,r]	6.16	Estimates minutes of advertising required for 3,000 in sales.

Weighted Mean

The following procedure calculates the weighted mean of data points $x_1, x_2, \ldots, x_n$ occurring with weights $y_1, y_2, \ldots, y_n$.

1. Use [INPUT] and [Σ^+] to enter x,y pairs. The y-values are the weights of the x-values.
2. Press [→] [6 $\bar{x}_{w,b}$].

Example 5

A survey of 266 one-bedroom rental apartments reveals that 54 of them rent for 500 per month, 32 for 505, 88 for 510, and 92 for 516. What is the average monthly rent?

Table 12-13 Calculating the average monthly rent

Keys	Display	Description
[↓] [→M / C STAT]	**STAT CLR** (message flashes briefly, then disappears)	Clears statistics memory.
5 0 0 INPUT 5 4 Σ+	1.00	Enters first rent and its weight.
5 0 5 INPUT 3 2 Σ+	2.00	Enters second rent and its weight.
5 1 0 INPUT 8 8 Σ+	3.00	Enters third rent and its weight.
5 1 6 INPUT 9 2 Σ+	4.00	Enters fourth rent and its weight.
[↓] [6 / $\bar{x}_{w,b}$]	509.44	Calculates the weighted mean.

Regression Models and Variables

The 10bII+ has six built-in regression models, as well as the ability to calculate which model best fits the current data. These six regression modes are listed in the table below.

Table 12-14 Regression models

Number and Mode	Description
0-Best Fit	Automatically selects fit
1-Linear	$m*x+b$
2-Logarithm	$m*\ln(x)+b$
3-Exponential	$b*e^{(m*x)}$
4-Power	$b*x^m$
5-Exponent	$b*m^x$
6-Inverse	$m/x+b$

Press [↑] [REGR / 1] to open the regression selection application. The initially displayed option is the current setting. Press [+] or [−] to scroll through the available regressions. With the desired model displayed, press [INPUT] to select it. To exit without changing the current model, press [C]. As an alternative to scrolling, and if you know the number of the desired model, press [↑] [REGR / 1] followed by the desired number of the fit option.

If **BEST FIT** is selected, the 10bII+ calculates the best fit when [↴] [4 $\hat{x}$,r], [↴] [5 $\hat{y}$,m], or [↴] [6 $\bar{x}_{w,b}$] is pressed. When selected, **BEST FIT** flashes briefly, followed by the chosen fit. The selected regression will remain set until a new one is selected, or the statistics memory is cleared.

When the statistics memory is cleared using [↴] [→M C STAT], the current regression model is set back to **LINEAR**.

Probability Calculations

In many probability calculations, specific methods of counting possible outcomes are required as part of a process to determine the likelihood of certain results. The three main operations that allow this are:

- ***!*** factorial
- $_{n}P_{r}$ permutations
- $_{n}C_{r}$ combinations

Factorial

Factorial (*!*) is a mathematical operator that instructs you to multiply the current number by all previous whole numbers. Writing out so many numbers can be cumbersome, so mathematicians use *!* to signify this process. For example:

5! is equivalent to 5 **x** 4 **x** 3 **x** 2 **x** 1 = 120.

On the 10bII+, the input value ***n*** must be within -253 < ***n*** < 253. The gamma function is used to calculate ***n***! for non-integer or negative values.

Permutations

The ***n***Pr function calculates the number of different arrangements, or permutations, of ***n*** items taken ***r*** at a time. No item can occur more than once in a set of ***r*** items, and different orders of the same ***r*** items are counted separately. This is calculated using the formula:

$$PERMUTATIONS = \frac{n!}{(n-r)!}$$

Example

Using five books labeled A, B, C, D, and E, how many different ways can three books be placed on a shelf?

Table 12-15 Example calculating permutations

Keys	Display	Description
5 [2nd] nPr 3 =	60.00	Calculates permutations of n items taken r at a time.
or, using [INPUT]:		
5 [INPUT] 3 [2nd] nPr	60.00	

Combinations

The nCr function calculates the number of different sets, or combinations, of n items taken r at a time. No item can occur more than once in the set of r items, and different orders of the same r items are not counted separately. This is calculated using the formula:

$$COMBINATIONS = \frac{n!}{(n-r)!\,r!}$$

Example

Using five colored balls, how many different color combinations of three colors can be chosen?

Table 12-16 Example calculating combinations

Keys	Display	Description
5 [2nd] nCr 3 =	10.00	Calculates combinations of n items taken r at a time.
or, using [INPUT]:		
5 [INPUT] 3 [2nd] nCr	10.00	

Random Number and Seed

The 10bII+ includes a random number generator function that generates a pseudo-random number in the range $0 < x < 1$. To store a seed value, type a positive number and press [RCL] [STO] [2nd] [RAND]. Storing a value of 0 will select a random number and store it as the seed value.

Example

Store a seed value of 42; set the number display to 9. Then generate three random numbers.

Table 12-17 Example storing a seed value and generating random numbers

Keys	Display	Description
4 2 ↘ RCL STO ↗ RAND =	42.00	Stores 42 as the random number generator seed.
↘ = DISP 9	42.000000000	Set display precision.
↗ RAND =	.199873749	Generate first random number.
↗ RAND =	.863046890	Generate second random number.
↗ RAND =	.504024868	Generate third random number.
↘ = DISP 2	.50	Reset display to default setting.

Advanced Probability Distributions

The 10bII+ allows easy calculation of the Z and Student's T probability distribution values. In addition, it allows inverse calculations of both functions. The values are calculated using the lower tail probability. This lower tail probability corresponds to the area under the curve to the left of the input. If you need a value other than a lower tail, such as a two-sided value, please see the conversion instructions at the end of this chapter.

Table 12-18 Advanced probability keys

Keys	Description
↗ Z⇄P 3	Calculates a cumulative normal probability given a Z-value.
↗ INV M+ Z⇄P 3	Calculates a Z-value given a cumulative normal probability.
↗ df,t⇄P 2	Calculates the cumulative Student's T probability given degrees of freedom and a T-value.
↗ INV M+ df,t⇄P 2	Calculates a T-value given degrees of freedom and the cumulative Student's T probability.

These distribution functions replace the statistical tables found in the back of textbooks. Unlike the textbook, the calculator can calculate any value, not just a limited selection found in the table.

Normal Lower Tail Probability

To calculate the area under the curve to the left of *z* (the lower tail probability), enter the *z*-value and press [SHIFT] [Z⇄P] (3). This function calculates the probability that a standard normal random variable, Z, is less than *z*.

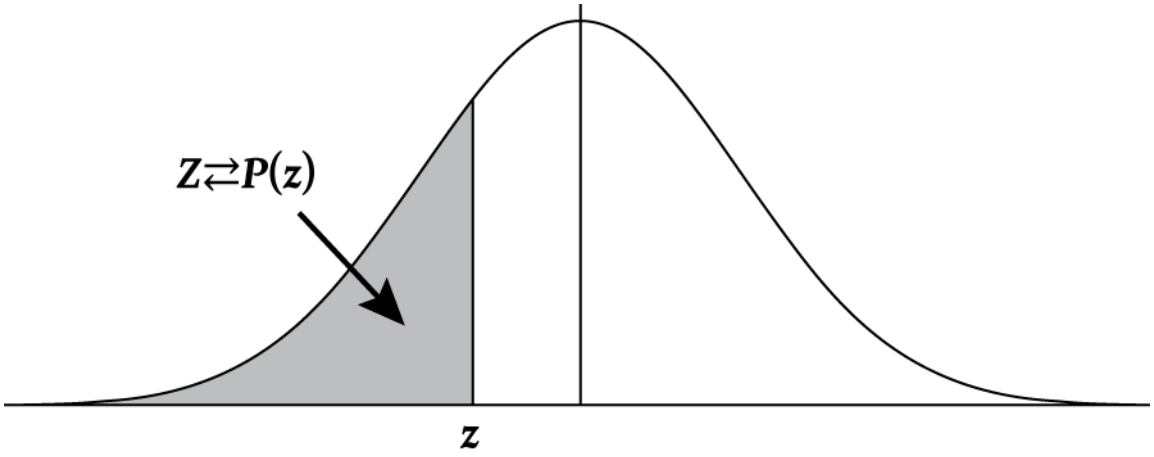


Figure 5

Example

The variable Z is a standard normal random variable. What is the probability that Z is less than -1.7.?

Table 12-19 Probability example

Keys	Display	Description
[SHIFT] [=] (DISP) [6]	.000000	Set display precision.
[1] [.] [7] [+/−] [SHIFT] [Z⇄P] (3)	.044565	Calculate the probability.

z	.00	.01
-1.90	.0287	.0281
-1.80	.0359	.0351
-1.70	.0446	.0436
-1.60	.0548	.0537

Figure 6

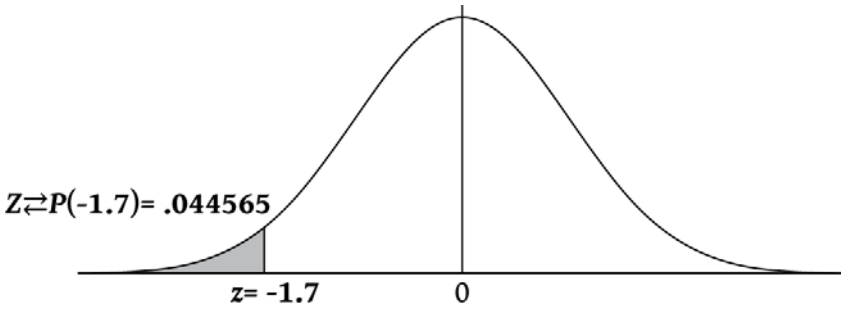


Figure 7

Inverse of Normal Lower Tail Probability

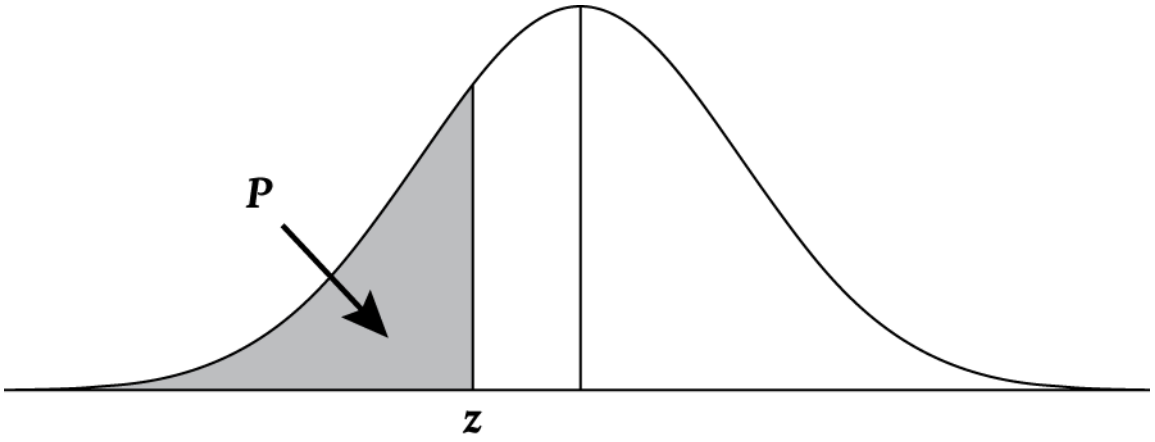


Figure 8

What is the *z*-value corresponding to a lower tail cumulative probability of .025?

Table 12-20 Example calculating z-value (lower tail)

Keys	Display	Description
[•] [0] [2] [5] [2nd] [INV M+] [Z⇄P 3]	-1.959964	Calculate the corresponding z-value.

z	.05	.06
-2.00	.0202	.0197
-1.90	.0256	.0250
-1.80	.0322	.0314
-1.70	.0401	.0392

Figure 9

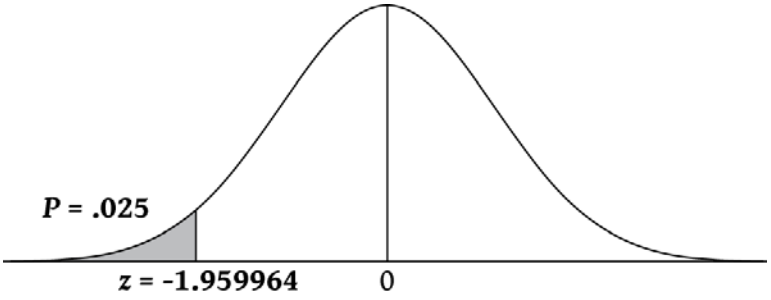


Figure 10

Student's T Probability Lower Tail

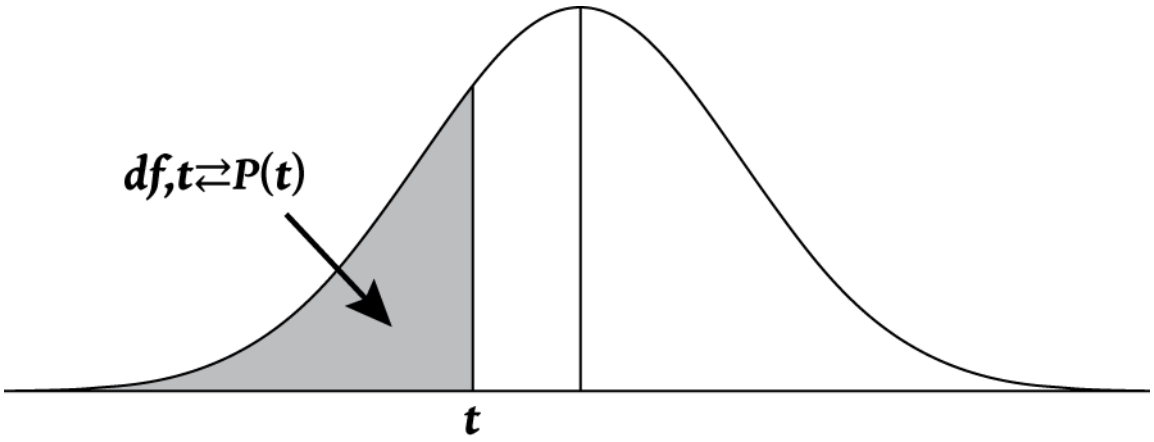


Figure 11

To calculate the area under the Student's T Distribution curve, first enter the degrees of freedom, followed by the *t*-value. It is a two-number function, so it may be entered as either an in-line function, or by using INPUT.

Example:

What is the lower tail probability associated with a Student's T distribution with 8 degrees of freedom (df_1) with a *t*-value of -1.86?

Table 12-21 Example of Student's t (lower tail)

Keys	Display	Description
8 2nd df,t⇄P 2 1 . 8 6 +/– =	.0499653	Calculates the lower tail probability.
or, using INPUT:		
8 INPUT 1 . 8 6 +/– 2nd df,t⇄P 2		Returns the same results.

df	.05	.01
6	-1.943	-3.143
7	-1.895	-2.998
8	-1.860	-2.896
9	-1.833	-2.861

Figure 12

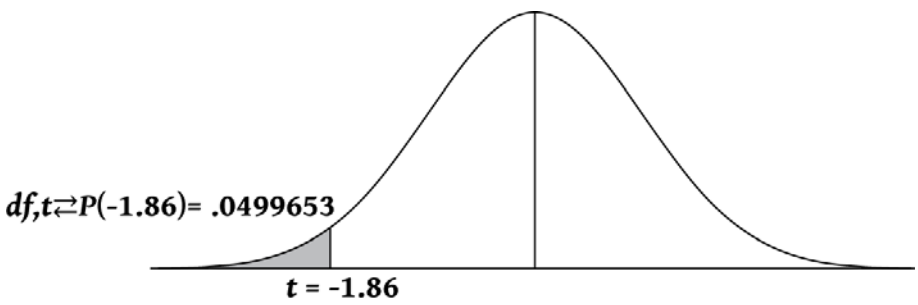


Figure 13

Inverse of Student's t Probability Lower Tail

If you know the lower tail probability, P, and you want to calculate ***t***, enter the degrees of freedom (df_1), followed by [INPUT], then P. Press [2nd] [INV] [df,t⇄P] [2] to calculate ***t***.

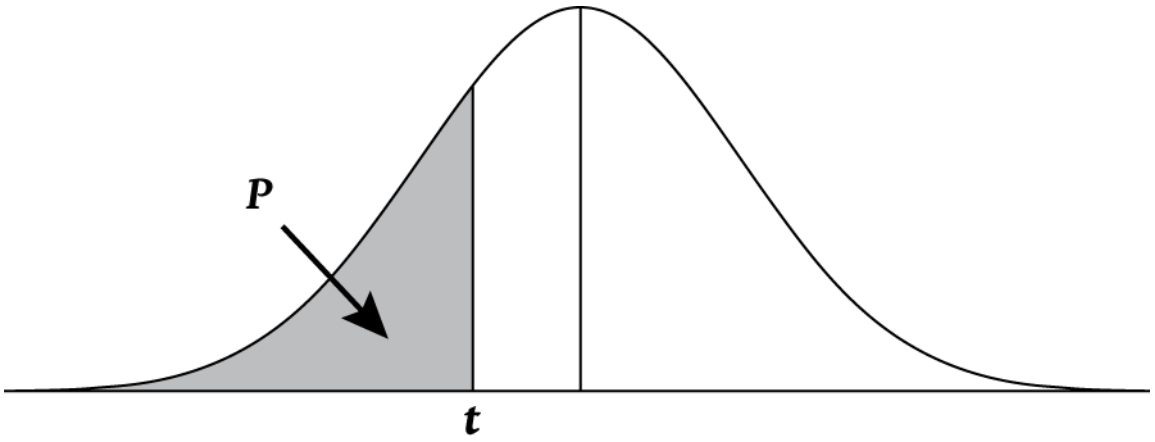


Figure 14

Example

A hypothesis test requires a critical ***t***-value from the Student's T distribution with 26 degrees of freedom. Find the ***t***-value for a lower tail probability of .05.

Table 12-22 Example calculating the *t*-value (lower tail)

Keys	Display	Description
2 6 [2nd] [INV] [df,t⇄P] [2] . 0 5 =	-1.705618	Enter degrees of freedom and the probability, and calculates the lower tail t- value.
or, using [INPUT]:		
2 6 [INPUT] . 0 5 [2nd] [INV] [df,t⇄P] [2]		Returns the same results.

df	.05	.01
24	-1.711	-2.492
25	-1.708	-2.485
26	-1.706	-2.479
27	-1.703	-2.473

Figure 15

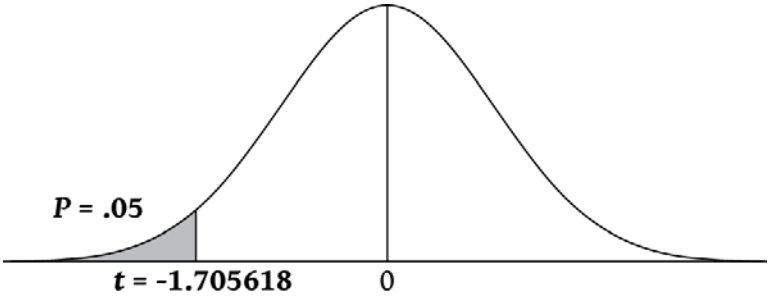


Figure 16

Conversions from Lower Tail

The distribution functions on the 10bII+ return values for the lower tail cumulative probability. The lower tail probability corresponds to the area under the curve to the left of the given value. Sometimes you will want to work with areas other than the lower tail. It is easy to convert from lower tail to another area as long as you keep in mind that the total area under the curve is equal to 1, and the Normal and the Student's T distributions are symmetrical. In other words, the portion of the curve to the left of zero is a mirror image of the portion of the curve to the right of zero.

Example 1

The random variable Z is a standard normal random variable. What is the probability that *z* is greater than -1.7?

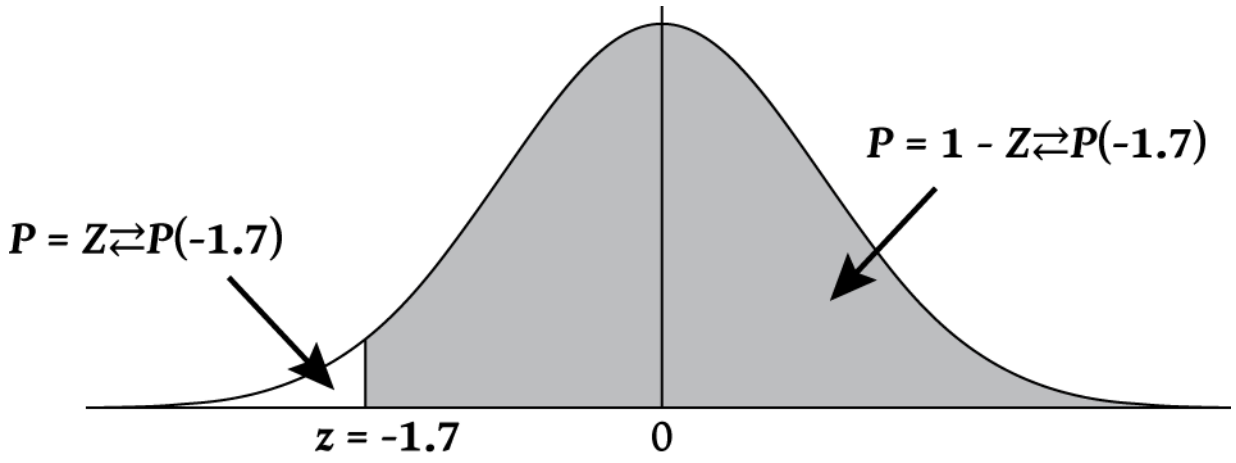


Figure 17

The probability that *z* is greater than -1.7 is the area of the curve to the right of -1.7. You can calculate the area to the left of -1.7 and subtract it from 1 (total area of the curve).

Table 12-23 Example converting from lower tail

Keys	Display	Description
1 • 7 +/– SHIFT Z⇄P (3)	.044565	Calculate the lower tail area. Since the area is -1.7, change the sign.
+/– + 1 =	.955435	Subtracts the lower tail from 1.

Example 2

The variable Z is a standard normal random variable. What is the probability that *z* is greater than 1.2 or less than -1.2?

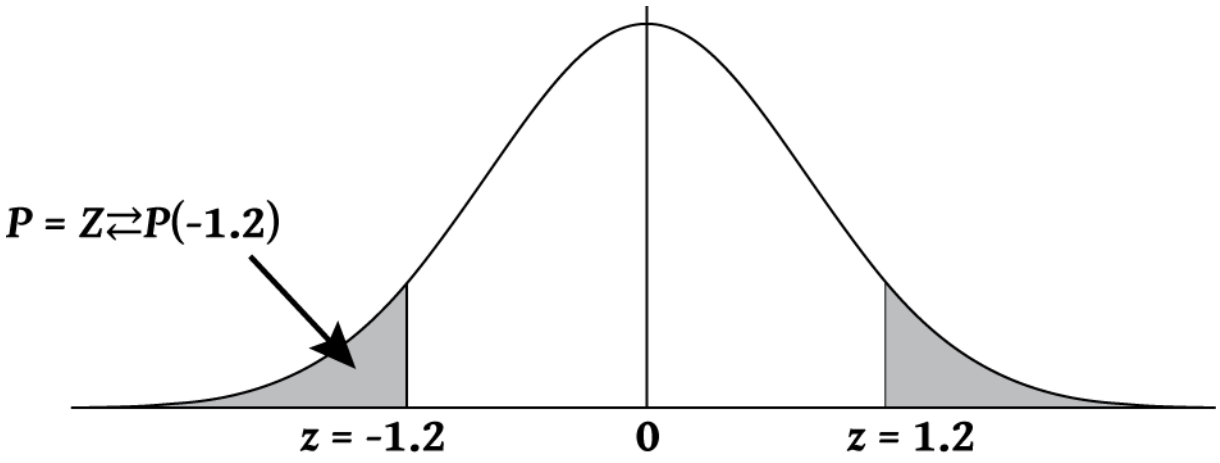


Figure 18

The desired area is to the right of 1.2 and to the left of -1.2. Since normal distributions are symmetrical, and the areas are the same, you can calculate the lower tail area and simply multiply by 2.

Table 12-24 Example converting from lower tail

Keys	Display	Description
1 . 2 +/– → Z⇄P (3)	.115070	Calculate the lower tail area and store the value.
× 2 =	.230139	Calculates the result.

Example 3

The variable Z is a standard normal random variable. Find *z* so that the probability that Z is less than *z* and greater than -*z* is equal to 0.95.

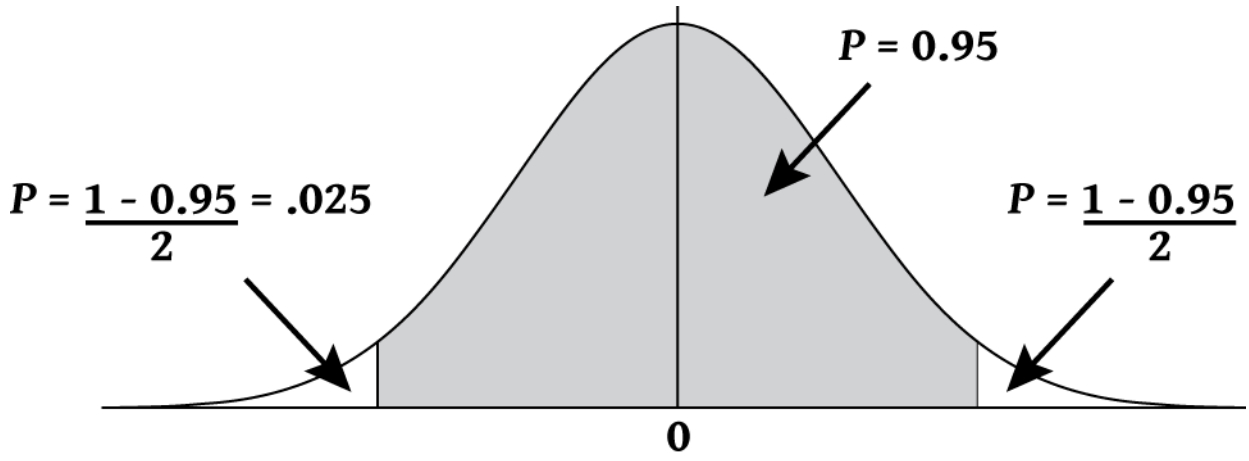


Figure 19

The given area is 0.95. The area not included is 1-0.95/2 = 0.025. Since the normal distribution is symmetrical, half of the desired area is in the lower tail, .05/2=.025. The desired area corresponds to a lower tail probability of .025.

Table 12-25 Example converting from lower tail (the inner area)

Keys	Display	Description
[.] [0] [2] [5] [+/-] [INV] [M+] [Z⇄P] [3]	-1.959964	Returns desired value of *z*.